Significant effects of frontal stimulation on sleep	**Direct current**	**Alternating current**
Sleep oscillations		
NREM stimulation	↑ **So** (Marshall, 2004)	↑ **So** (Cellini, 2019; Marshall, 2006; Munz, 2015; Passmann, 2016; Saebipour, 2015; Westerberg, 2015)
	↑ **β** (Frase, 2016)	↑ **Sp** (Ladenbauer, 2017; Marshall, 2006; Passmann, 2016)
Post-stimulation	↑ **γ** (Frase, 2016)	↓ **Δ θ** (Eggert, 2013; Garside, 2015)
		↑ **So, Sa, Sp** (Koo, 2015; Marshall, 2011; Passmann, 2016; Reato, 2013)
Sleep patterns		
NREM stimulation	↓ **SE** (Roizenblatt, 2007)	↑ **N2 N3** (Eggert, 2013; Ladenbauer, 2017; Marshall, 2011) ↓ **N1** (Cellini, 2019)
Post-stimulation	↑ **REM** (Roizenblatt, 2007) ↑ **SOL** (Roizenblatt, 2007)	↓ **N2 N3** (Marshall, 2011; Sahlem, 2015)
Entire stimulation night	↓ **SE** (Frase, 2016) ↓ **TST,** ↑ **WASO** (Frase, 2016)	↑ **N2 N3 SWS** (Cellini, 2019; Marshall, 2006) ↓ **N1** (Cellini, 2019) ↓ **N4** (Passman, 2014) ↓ **SE** (Johnson and Durrant, 2018)
Subjective effects		
Subjective sleep improvement	↑ **subjective parameters** (Sheng, 2018).	↑ **subjective parameters** (Charest, 2019; Robinson, 2018)

Supplementary Material S2: Highlights of major significant effects of frontal tES on sleep with corresponding references. Results regarding non-frontal tES, tES during REM and fast alternating tES were scarce, and thus omitted from the Table.

Effect on sleep patterns: NREM = non rapid eye movement sleep; N1-4 = NREM sleep stages relative duration; REM = rapid eye movement sleep; SE = sleep efficiency; SWS = slow wave sleep relative duration; SOL = sleep onset latency; WASO = wake after sleep onset.
Effects on sleep oscillations: β = beta power spectral density PSD (15-25 Hz); Δ = delta PSD (1-4 Hz); γ = gamma PSD (> 25 Hz); θ = theta PSD (4-8 Hz); NREM = non-rapid eyes movement sleep; So = slow oscillations PSD (0.5-1 Hz); Sa = slow activity PSD (0.5-4 Hz); Sp = spindle PSD (10-15 Hz).